

SORTWISE - Deliverable D3.2
Pilot results of NIR sorting lines
Work package 3 - Public - 15 March 2024

Executive summary

Two pilot sorting lines combined near-infrared (NIR) spectroscopy with colour cameras to sort post-consumer textiles by fibre type and colour. Over 18 months the lines sorted 1,450 tonnes. Fibre purity of the cotton fraction reached 96 percent and of the polyester fraction 93 percent at a throughput of 900 kg per hour.

Results per fraction

Cotton (white and light colours): 31 percent of input, sold to a mechanical recycler.

Polyester: 18 percent, sent to chemical recycling trials.

Blends: 27 percent, no stable outlet yet.

Wool: 4 percent, sold to a Prato spinner.

Data offered

The anonymised spectra library (12,000 labelled garments) and the monthly fraction statistics are published as an open dataset for sorters and recyclers.